

LMS Backend — Project Report

تقرير مشروع نظام إدارة التعلم — اللي تعمل لحد الوقتي

Laravel 13 · MySQL · Modular monolith (nWidart) · JWT + Spatie Teams

Phases 1–3 + pre-Phase 7 refinement (ADR-011–015)

Repository: github.com/AhmedTyson/lms-laravel-backend · September 2026

Part 1 — Egyptian Arabic: what was built and why | Part 2 — English: full database schema | Part 3 — every rule: what it requires and what we implemented exactly.

الجزء الأول: لبنى لحد لوقتى — بالعامية

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المرحلة الأولى: لتأسيس

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المرحلة الثانية: لهيكل

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المرحلة الثالثة: قاعدة بيانات

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النتيجه قبل المرحلة السابعة

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Part 2 — Database Schema (English)

20 tables (19 spec + **user_permissions** read model, ADR-012). Auto-increment integer PKs everywhere (ARCH-005). Portable SQL: runs on SQLite (dev) and MySQL (prod). Conventions: **UQ** = unique, ***** = NOT NULL.

users (*Auth ext: manager_id, approval_status, manager_depth*)

Column	Type / constraints
id	bigint PK *
name	varchar(255) *
email	varchar(255) * UQ
email_verified_at	timestamp
password	varchar(255) *
manager_id	FK → users NULL (RULE-002)
manager_depth	smallint default 0 (ADR-013 cache)
approval_status	pending approved NULL (RULE-010)

Population registry. One manager per user = Admin-rooted tree; students always NULL.

groups (*owner_id → users; name NOT unique RULE-011*)

Column	Type / constraints
id	bigint PK * (= Spatie team_id)
owner_id	FK → users *
name	varchar(255) *

Permission scopes (“rooms”), not ranks. Ownership succession on removal (RULE-012).

group_members *UQ(group_id, user_id)*

Column	Type / constraints
id	bigint PK *
group_id	FK → groups *
user_id	FK → users *
joined_at	timestamp *

Guest list only — membership grants zero capabilities (RULE-009).

permission_grants *append-only, no updated_at (RULE-005)*

Column	Type / constraints
id	bigint PK *
granter_id	FK → users * RESTRICT
grantee_id	FK → users * RESTRICT
group_id	FK → groups NULL (=global)
permission_name	varchar *
action	granted revoked *

granted_at / revoked_at	timestamps
created_at	timestamp *

Court record of delegation: subordinate-only (RULE-003) + ceiling (RULE-004), explicit team context (ADR-008).

user_permissions #20, UQ(user, permission, group) — ADR-012

Column	Type / constraints
id	bigint PK *
user_id	FK → users * cascade
permission_name	varchar *
group_id	FK → groups NULL
granted_via_grant_id	FK → permission_grants *

O(1) authorization read model: CURRENT grants only, mirrored atomically with the ledger.

courses UQ(instructor, title)

Column	Type / constraints
id	bigint PK *
instructor_id	FK → users * RESTRICT
title / category	varchar *
description	text
status	draft published archived *
published_at / archived_at	timestamps

Linear lifecycle Draft→Published→Archived (RULE-013); archive blocked with active enrollments.

lessons UQ(course_id, order) RULE-014

Column	Type / constraints
id	bigint PK *
course_id	FK → courses * cascade
title	varchar *
content_reference	text (external pointer)
order	uint *

Numbered chapters; progress references IDs so reorder never corrupts.

enrollments UQ(student, course)

Column	Type / constraints
id	bigint PK *
student_id	FK → users *
course_id	FK → courses * cascade
status	active completed *

enrolled_at / completed_at	timestamps
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One seat per student per course; unique closes the concurrent-enroll race at DB level.

assignments *variable scoring v2.1*

Column	Type / constraints
id	bigint PK *
course_id	FK → courses * cascade
title	varchar *
description	text (doubles as instructions)
due_date	timestamp * required+future
resubmission_allowed	bool default false
max_score	decimal(6,2) * free-form
passing_threshold	decimal(5,2) * % of max

Free-form ceilings (any max); passing = % of own max (RULE-018 rev).

submissions *UQ(assignment, enrollment, attempt) RULE-015*

Column	Type / constraints
id	bigint PK *
assignment_id / enrollment_id	FKs * cascade
attempt_number	uint *
file_path / content	both nullable
submitted_at / is_late	timestamp / bool *
grade / graded_by / graded_at	nullable
status	submitted graded *

All attempts preserved, independently gradable — no grade-of-record (RULE-016).

quizzes *threshold = % of live point sum*

Column	Type / constraints
id	bigint PK *
course_id	FK → courses * cascade
title	varchar *
opens_at / closes_at	timestamps * (window at start)
max_attempts	uint default 1
passing_threshold	decimal(5,2) *

Timed exam room; threshold recomputed from questions, never cached.

questions *per-question points v2.1*

Column	Type / constraints
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id	bigint PK *
quiz_id	FK → quizzes * cascade
prompt	text *
type	5 auto-scorable types *
order	uint *
points	decimal(6,2) default 1.00
correct_boolean / correct_number / tolerance	type-specific, nullable

Weighted scorable moments; short answers accept variants (RULE-017).

question_options

Column	Type / constraints
id	bigint PK *
question_id	FK * cascade
label	varchar *
is_correct	bool *
order	uint *

Pre-marked choices; grading = comparison.

question_accepted_answers

Column	Type / constraints
id	bigint PK *
question_id	FK * cascade
answer_text	varchar *

Accepted phrasings for short answers (RULE-017).

attempts UQ(quiz, enrollment, attempt)

Column	Type / constraints
id	bigint PK *
quiz_id / enrollment_id	FKs * cascade
attempt_number	uint *
score	decimal NULL
started_at / submitted_at	timestamps

One sitting per row; NULL score = in progress.

attempt_answers UQ(attempt, question)

Column	Type / constraints
id	bigint PK *

attempt_id / question_id	FKs * cascade
selected_option_ids	json NULL
answer_boolean / text / number	nullable
is_correct	bool *

Per-question verdicts enabling review and appeals.

progress_records PK = enrollment FK (1-to-1)

Column	Type / constraints
enrollment_id	FK PK *
percent_complete	decimal default 0 *
completed_at	timestamp

Dashboard percentage cache; all components done $\Rightarrow$ course done.

component_completions UQ(enrollment, type, id); morph (ADR-010)

Column	Type / constraints
id	bigint PK *
enrollment_id	FK * cascade
component_type	lesson assignment quiz *
component_id	bigint * (no cross-FK)
completed_at	timestamp
is_override	bool default false (RULE-019)
overridden_by	FK $\rightarrow$ users NULL

Gold stars via \$completion->component; orphan-purged by observer (ADR-015).

notifications idx(user, read_at)

Column	Type / constraints
id	bigint PK *
user_id	FK * cascade
type	varchar * (domain event)
data	json *
read_at	timestamp NULL

Queued mailbox; users see only their rows.

activity_log package-owned (ADR-006); idx(created_at) (ADR-014)

Column	Type / constraints
id	bigint PK *
log_name / description	varchar/text
subject_* / causer_*	polymorphic, nullable

event / properties / batch_uuid	nullable
created_at / updated_at	timestamps

CCTV trail; pruned nightly after 90 days. Grants stay the court record.

Part 3 — Every Rule: what it requires, what we implemented

كل قاعدة تحت: معناها بالإنجليزية، بعدين سطر بالعامة يبقو عملنا فيه فيها بالظبط.

RULE-001 — Published-content edits need elevated permission

Requires: Editing a published course structure requires a permission instructors lack by default.

Implemented: Policy check lands in Phase 6 course slice; schema ready (status column).

المحتوى المنشور عقد — تعديله محتاج صلاحية خاصة، هه هيتطبق في مرحلة الكو سا.

RULE-002 — Strict single-manager tree rooted at Admin

Requires: Every user has exactly one manager_id forming a tree; students always NULL.

Implemented: Migration 120000 adds manager_id self-FK + index; seeder sets admin NULL; Admin-first contract tested.

كل حد له مدير حد بس، لشجر جذها لا من. تنفذ في قاعدة بياناتنا لسيدي.

RULE-003 — Subordinate-only delegation

Requires: Grants/revokes only to direct or indirect subordinates.

Implemented: Enforced in Phase 5 GrantService; test matrix defined in Task 5.2 (403 otherwise).

محدد يد صلاحية لحد مش تحته في لشجر — هيتطبق في مرحلة لصلاحية.

RULE-004 — Ceiling rule

Requires: You can only grant permissions you personally hold (global or group-scoped).

Implemented: Checked with explicit team context per call (ADR-008); manager_depth cache (ADR-013) accelerates reads.

متدريش حاجة مش معاك — لقاعد قلب نظا لتفويض.

RULE-005 — Append-only audit ledger

Requires: Every grant AND revoke permanently recorded; nothing updated or deleted.

Implemented: permission_grants has no updated_at; revoke = new row; prune command explicitly excludes it.

سجل لا يتمسح: كل منح مسح بيتسجل للأبد.

RULE-006 — Reassignment is non-retroactive

Requires: Changing a manager never rewrites history granted by the old manager.

Implemented: Service rule for Phase 5; history rows carry no cascade that could rewrite them.

تغيير للمدير مش يغير لماضي.

RULE-007 / RULE-008 — Global vs group-scoped permissions

Requires: A grant is either global (group_id NULL) or valid inside one group.

Implemented: Nullable group_id FK; user_permissions mirrors both shapes with lineage.

لصلاحية يا عامة يا جو جر حد بس.

RULE-009 — Membership ≠ permission

Requires: Joining a group grants zero capabilities.

Implemented: Separate group_members table; test asserts member-without-grant gets 403 (Task 5.3).

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RULE-010 — manager_id set at approval, not registration

Requires: Instructor manager assigned when Admin approves, not when they register.

Implemented: Phase 4 approval endpoint wires ManagerAssignmentService; column nullable until then.

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RULE-011 — No group-name uniqueness

Requires: Group names may repeat; groups are scopes, not identities.

Implemented: No unique index on groups.name (migration 120100).

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RULE-012 — Group ownership succession

Requires: Removed owner → ownership passes to their manager, else any Admin.

Implemented: Phase 5 Task 5.3 with succession tests; tie-break deferred per spec.

ﺍﻟﻤﺎﻟﻚ ﻟﻮ ﻣﺶ، ﺍﻟﻤﻠﻜﻴﺔ ﺑﺘﺮ ﺍﻟﻤﺪﻳﺮﻩ ﺍﻟﺪﺭﺱ ﻟﺂ ﺍﻟﺪﺭﺱ.

RULE-013 — Linear course lifecycle

Requires: Draft → Published → Archived only; archive blocked with active enrollments.

Implemented: status column + default draft; transitions enforced in Phase 6 service with tests.

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RULE-014 — Lesson order unique per course

Requires: Two lessons in one course can never share a position.

Implemented: Composite unique (course_id, order); duplicate-order test proves it fires.

ﺗﺮﺗﻴﺐ ﺍﻟﺪﺭﺱ ﻓﺮﻳﺪ، ﺍﻟﻘﺎﻋﺪﺍﺕ ﺍﻟﺠﻴﺎﻧﺎﺕ ﻧﻔﺴﻬﺎ ﻳﺘﻤﻨﻊ ﺍﻟﺘﻜﺮﺭ.

RULE-015 — Resubmission + race-safe attempts

Requires: Second attempts only if allowed; attempt_number unique stops race duplicates.

Implemented: Composite unique on submissions + attempts; concurrent-submit test in Phase 8/9 tasks.

ﻣﻔﻴﺶ ﻣﺤﺎﺗﻠﺔ ﺗﺎﺑﻴﺔ ﻟﻮ ﻣﺴﻤﻮﺍﺕ، ﺍﻟﺘﻠﺴﺒﺎﻗﺎﺕ ﻣﻘﻮﻟﺔ ﺑﻔﻬﺮ ﻓﺮﻳﺪ.

RULE-016 — No grade-of-record

Requires: All attempts preserved and independently gradable.

Implemented: No summary column exists by design; grading writes per-attempt grade only.

ﻛﻞ ﺍﻟﻤﺤﺎﺗﻼﺕ ﻣﺤﻔﻮﺿﺔ ﻣﻔﻴﺶ ﺍﻟﺘﺠﺔ ﺍﻟﺪﺭﺱ ﺑﺘﻤﺴﺢ ﺍﻟﺒﺎﻗﻲ.

RULE-017 — Multi-variant short answers

Requires: Short-answer questions accept several correct phrasings.

Implemented: question_accepted_answers table; scoring compares against all variants (Phase 9).

RULE-018 (revised v2.1) — Percentage thresholds over variable totals

Requires: Completion = best attempt $\geq$ threshold % of that component's own max (not 0–100).

Implemented: questions.points, quizzes.passing_threshold, assignments.max_score+passing_threshold columns; Quiz::totalPoints() sums live.

النسبة من مجموع كل مكون على حد، مش من 100 ثابتة.

RULE-019 — Instructor override

Requires: Instructor/Admin may manually mark completion, attributed.

Implemented: is_override + overridden_by columns; override path in Phase 10 engine.

المدد يقدم ينح حد يد، بس باسمه متسجل.

SCOPE-001 — Payments are future scope

Requires: Out of scope now.

Implemented: Nothing payment-shaped exists anywhere.

لدفع مش لوقتي.

SCOPE-002 — Access Management is a separate module

Requires: Delegation is a general subsystem, not teaching logic.

Implemented: Own AccessManagement module: groups, grants, read model, depth helpers.

لصلاحيات موليوي مستقل.

SCOPE-003 — Groups are first-class scoped entities

Requires: Groups exist as permission scopes.

Implemented: groups + group_members tables; Spatie team_id = group id.

لجرها كيانا أساسية.

SCOPE-004 — Instructor approval workflow

Requires: Instructors browse-only until Admin approves.

Implemented: approval_status column; Phase 4 approval endpoint + event.

المدد يستنى مولفقة لآمن.

SCOPE-005 — Five auto-scorable question types

Requires: single/multi choice, true/false, short-answer, numeric.

Implemented: type column + per-type answer columns; ScoringService in Phase 9.

خسة لولي مسئلة بتصح لي.

ARCH-005 — Auto-increment IDs everywhere

Requires: No UUIDs, no exceptions.

Implemented: Every migration uses \$table->id(); platform pin keeps lock consistent.

قافا مسلسلة في كل لجد.

ARCH-CONSTRAINT-002 — Mandated stack

Requires: Tymon JWT, Spatie Teams, Redis, Activitylog, Scramble, Pest, Pint.

Implemented: All installed with L13-compatible versions recorded in README/AGENTS.md.

الباك دجاجة المطلوبة كلها مكتبة.

ADR-006 — activity_log formalized as table #19

Requires: Package-owned table documented, not mistaken for drift.

Implemented: Spatie publish untouched; reproduced in spec §5.20.

جدد تعلم رقبة بنات الباك دجاجة متوثق.

ADR-007 — Service-layer cycle prevention

Requires: No DB constraint can police a tree; transactional service check instead.

Implemented: Rule locked; ManagerAssignmentService arrives in Phase 5 with FOR UPDATE + dedicated exception.

منع الحلقا خدمة مش يقيد قاعدة بيانات.

ADR-008 — Spatie Teams context split

Requires: Ambient for single-group flows; explicit per-call for grants + reporting.

Implemented: Lint-able convention; stale-context regression tests required in Tasks 5.2/12.1.

سيا لفريق صريح في الحسا، اضميني في لعا.

ADR-009 — Admin seeder first

Requires: Exactly one root Admin from .env, manager NULL, before all else.

Implemented: DatabaseSeeder firstOrCreate + Schema guard; verified seeding 1 admin; never hardcoded.

لا لمن لا من ملف البيئة.

ADR-010 — Polymorphic completions

Requires: morphTo/morphMany, no cross-table FK; code-layer only.

Implemented: ComponentCompletion::component() + morph map + completions() on all three; SchemaTest proves resolution.

علاقة مرنة بد مفاتيح مستحيلة.

ADR-011 — Explicit RESTRICT on user-owned FKs

Requires: Course/grant owners cannot be hard-deleted; soft-delete only.

Implemented: Migration 120600 restrictOnDelete (MySQL; SQLite skips by design); delete-blocked tests green.

صاحب لكو سا لسا مبيتمسحش.

ADR-012 — user_permissions read model (table #20)

Requires: O(1) auth reads; ledger stays history source; atomic writes.

Implemented: Table + UserPermission model + PermissionReadModelSync seam tested; Phase 5 wires atomicity.

ADR-013 — manager_depth cache

Requires: Denormalized depth (root=0), subtree recompute in-transaction, no cap.

Implemented: Column + ManagerDepth helper tested on 3-level tree; Phase 5 service calls it post-cycle-check.

عمق 222 لشجر 222 متخز 222 للسرعة 222 ليتحد 222 مع كل نقل.

ADR-014 — 90-day activity_log retention

Requires: Nightly prune via indexed created_at; grants never pruned.

Implemented: Migration 120603 index + PruneActivityLog command + daily schedule + env var; both prune tests green.

سجل 222 لمر 222 قبة يتنصف كل 90 يو 222.

ADR-015 — Orphan protection without FK

Requires: Observer purges completions when components die; trashed-keep rule for future.

Implemented: ComponentDeletionObserver on all three models + deleted()/forceDeleted() semantics; 2 purge tests green.

مر 222 قبة يمسح 222 لا 222 لما 222 ليتيم 222 لما 222 لمر 222 يمسح.

Verification & sign-off

Gates run in order, all green: **Pint** passed · **PHPStan** (Larastan wired) no errors · **migrate:fresh** 28 migrations · **Pest** 20 passed / 78 assertions (9 RefinementTest + 11 existing).

CI mirrors the same four checks on PHP 8.4. Phase checkpoints awaiting human sign-off before Phase 4 (Auth slice).

كل ☐ لبطون ☐ خضر ☐ للتنسيق ☐ لتحليل ☐ لترحيل ☐ لاختبار ☐ للمشر ☐ جاهز لمرحلة ☐ API.